

Electrical Bench Layout + Test Guide

As-of date: 2026-05-27

Purpose: provide a practical, print-friendly game plan for building and validating the electrical module mechanically before final wiring, energization, or permanent enclosure/panel closeout.

Related docs:

- docs/core/ELECTRICAL_48V_ARCHITECTURE.md
- docs/implementation/ELECTRICAL_overview_diagram.md
- docs/implementation/ELECTRICAL_fuse_schedule.md
- docs/plans/PROJECT_build_order_of_operations.md

1) Build intent lock (this guide assumes)

- Architecture remains 48V core + 12V distribution + shore/inverter AC .
- Alternator path baseline is the dedicated 48V secondary alternator path (Mechman + WS500 + APM-48).
- Electrical closet is a two-plane vertical corner build:
 - Board A:** against bed wall (high-current 48V hardware).
 - Board B:** against bulkhead (controls, 12V , AC support hardware).
- Closet sits above battery zone.
- Structural mounting intent is through-bolts with washers and lock nuts (no wood screws for permanent component mounting).
- The installed camper shell owns final outer dimensions and service-access geometry; garage/floor layouts are templates only until the two-plane mockup passes in-camper insertion, removal, cable-bend, fuse-access, and panel-access checks.

2) Recommended sequence (ground build -> test fit -> hard mount)

Current owner status (2026-05-27): the electrical module has passed the first live checkpoint. Owner confirmed 55.5V throughout the 48V system, MultiPlus inverter mode on with no errors, SmartShunt and Orion online in VictronConnect, Cerbo remote-console/AP workflow active, and a short limited-current shore-charge test completed.

Immediate priority:

- Preserve the current working topology; do not add alternator or AC-out branch complexity until cleanup checks pass.
- Add/verify labels, covers, torque witness marks, strain relief, cable protection, and service access around the energized layout.
- Program/verify the MultiPlus LiFePO4 charge profile with MK3-USB + VEConfigure or equivalent before sustained charging.
- Repeat shore-charge validation with logged voltage/current/SOC/temperature after charger settings are confirmed.
- Keep AC-out branch/GFCI validation and Mechman/WS500 alternator commissioning as separate later gates.

Phase L1: Build the two backer boards on the ground

- Rip the 1/2" sheet into Board A + Board B to your target envelope.
- Mark orientation on both boards: TOP , BOTTOM , FRONT , REAR , BED WALL , BULKHEAD .
- Place boards at 90 degrees on temporary braces/clamps to represent closet geometry.

Exit criteria:

- Two boards stand in correct orientation and are easy to reposition.

Phase L2: Paper layout before drilling

- Make 1:1 paper templates for each major component.
- On each template, mark:
 - footprint
 - hole centers
 - cable entry/exit sides
 - keep-out for bend radius + tool access
- Tape templates to boards and pencil trace candidate positions.
- Reserve explicit channels:
 - high-current 48V trunk path
 - low-current control/comms path

- AC route path separated from DC path

Exit criteria:

- No blocked fuse access, no blocked terminals, and no impossible cable bends.

Phase L3: Truck-bed dry fit (before full drilling)

1. Move bare boards (or cardboard stand-ins with same outer dimensions) into truck bed.
2. Validate insertion, rotation, and removal path.
3. Validate service access around door swing/panels.
4. Adjust board size or corner angle now if needed.

Exit criteria:

- Boards can be inserted and removed without forcing or trimming installed components.

Phase L4: Pilot drill + subset mount

1. Pilot drill only (1/8" class) for critical components first.
2. Mount a subset first (example: disconnect, Lynx, shunt, Orion, one AC enclosure).
3. Re-check wrench access, cable swing, and service pull paths.
4. Only after pass, drill final hole sizes and mount remainder.

Exit criteria:

- Critical devices fit and are serviceable with real tools/hands.

Phase L5: Final mounting + board prep

1. Through-bolt permanent component mounts.
2. Add backing washers/plates where loads are concentrated.
3. Seal plywood faces/edges after hole pattern is validated.
4. Label each component position and cable corridor directly on board.

Exit criteria:

- Boards are mechanically complete and labeled for wiring.

3) Suggested zoning by board

Board A (bed-wall plane, high-current focus)

- Place heavier/high-current gear lower when possible for stability and short battery paths.
- Keep these grouped for short 48V runs:
 - main disconnect
 - Lynx
 - shunt
 - major 48V feeds/returns
- Keep alternator branch landing (F-04 path) accessible and clearly labeled.
- Keep Class-T and major fuse service points reachable without removing adjacent equipment.

Board B (bulkhead plane, controls + distribution)

- Place 12V panel, Orion, monitoring/comms, and AC panel hardware here.
- Keep AC hardware grouped and physically separated from DC/control paths.
- Keep control and comms routing high/clean, away from high-current cable bundles.

4) Wiring layout rules (before you crimp anything)

- Keep parallel battery paths balanced by similar total loop resistance. Equal positive-only leads are not required if short/medium/long positives are offset by long/medium/short negatives.
- Keep 48V high-current paths short and direct.
- Keep AC and DC routing physically separated; cross at 90 degrees only when unavoidable.
- Use abrasion protection at every edge pass-through.
- Leave service loops at removable devices (do not strain to exact length).

- Do not cut final shell-dependent runs yet; only bench-module harnesses at this stage.

5) Bench test ladder (staged energization)

Use a stop/go ladder; do not skip steps.

T0: Dead checks

- Verify polarity labels and continuity on every cable.
- Verify torque marks applied to all completed terminations.
- Verify fuse IDs/ratings match `docs/implementation/ELECTRICAL_fuse_schedule.md`.
- Before any AC energization, meter the AC system dead: AC-in and AC-out neutrals are isolated, PE/equipment ground is continuous through inlet/enclosure/MultiPlus/receptacle paths, and no fixed downstream neutral-ground bond exists.

Pass when:

- No polarity/continuity mistakes and all terminations documented.

T1: Low-risk controls and comms

- Validate control circuits and monitoring links first (`WS500` control path, comms, panel labeling).
- Confirm `Upfitter #3` control path is treated as low-current signal only.

Pass when:

- Control/monitoring paths behave as expected with no unexplained faults.

T2: 48V core energization (no alternator)

- Bring up house `48V` path first.
- Use pre-charge before closing the main disconnect into MultiPlus/inverter capacitance.
- Verify expected voltage at key nodes and no abnormal heat/smell/noise.
- Confirm disconnect behavior and basic inverter/charger DC-side response.

Current result (`2026-05-27`):

- Owner measured `55.5V` throughout the system, including at the MultiPlus.
- MultiPlus switch `I` brought inverter mode online; inverter light illuminated, slight hum was observed, and no error lights were reported.
- SmartShunt and Orion-Tr Smart were visible in VictronConnect.

Pass when:

- Stable voltage and normal behavior at all checkpoints, with labels/covers/torque marks verified after the first live session.

T2.5: Shore / MultiPlus initial charge validation

- Build the AC-in-only path first: shore source/adapters -> portable EMS -> shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in.
- Keep AC-out branch breakers/loads disconnected for the first charge test.
- Confirm the T0 AC dead-checks are complete before shore energization.
- Set AC Input 1 label/type to `Shore power` and set the MultiPlus input current limit to the actual source. Use `10A` for first household tests and `12A` maximum policy on a normal `15A` outlet.
- Leave `DVCC` disabled unless a documented BMS/GX control path is added.
- Before sustained charging, program/verify MultiPlus LiFePO4 charge profile from the Dumfume manual (`MK3-USB` + `VEConfigure` or equivalent): absorption/float, charge current, equalization off, and lithium temperature-compensation behavior.
- Use the pre-charge procedure before closing the `48V` disconnect into MultiPlus capacitance.
- Leave alternator branch inactive and `F-04` out during shore-charge validation.
- Log starting/resting voltage, charge current, temperature, SOC/monitor readings, abnormal heat/noise/smell, input-current limit, and stop conditions.

Current result (`2026-05-27`):

- Short AC-in test passed at limited household-outlet draw: about `1294W` shore input and about `54.3V x 21.6A` (`~1173W`) battery charging in bulk.
- This is a functional test only; the MultiPlus battery profile still needs formal programming/verification before unattended or extended charging.

Pass when:

- Shore charging repeats normally with programmed battery settings, no unexplained BMS trips, abnormal heat, polarity issues, or AC faults, and logged values match expected limits.

T3: 12V distribution validation

- Validate Orion-fed 12V junction behavior and branch fusing.
- Test representative 12V loads one at a time, then combined.

Pass when:

- 12V rail remains stable under expected test loads.

T4: AC path validation

- Validate AC-in and AC-out paths in staged order with small known loads first.
- Reconfirm neutral isolation, PE continuity, no downstream neutral-ground bond, and correct outlet polarity before applying known loads.
- Confirm branch protection behavior and test each GFCI receptacle in its installed branch context.

Pass when:

- AC paths behave correctly in shore and inverter modes, with GFCI behavior verified and no neutral/ground faults found.

T5: Alternator path readiness (pre-engine integration)

- Confirm alternator branch hardware is present/labeled/serviceable.
- Confirm WS500 enable/disable control path labeling and fuse placement (F-15).
- Defer live engine alternator validation until full install context and safety controls are ready.

Pass when:

- Alternator branch is install-ready with shutdown path understood and labeled.

Stop conditions (halt and fix before continuing):

- Any reversed polarity, unexpected voltage, heating at terminals, damaged insulation, nuisance fuse trips, or unclear shutdown behavior.

6) Garage workday card (print/use)

Use one card per session:

- **Date:**
- **Board focus:** A / B / Both
- **Phase:** L1 / L2 / L3 / L4 / L5 / T0-T5
- **What changed:**
- **What passed:**
- **What failed or needs rework:**
- **Next single action:**

7) Hold points (do not cross yet)

- Do not finalize shell-dependent cable cuts.
- Do not run sustained/unattended MultiPlus charging until the charger profile is programmed/verified against the battery manual.
- Do not treat the short AC-in charge test as AC-out branch commissioning; GFCI/outlet validation remains separate.
- Do not lock roof/solar final terminations.
- Do not permanently close access panels before all staged tests pass.
- Do not use main 48V disconnect as first alternator shutdown method while charging is active; use Upfitter #3 (WS500 disable path) first.